

Michael Lin

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Education

Carnegie Mellon University

Pittsburgh, PA

M.S. in Computer Vision (Robotics Institute)

August 2025 – December 2026

- **Relevant coursework:** Machine Learning, Advanced Computer Vision, Learning for 3D Vision, SLAM, Robot Learning, Visual Learning and Recognition

Boston College

Chestnut Hill, MA

B.S. in Computer Science, B.A. in Mathematics

August 2021 – May 2025

- **Relevant coursework:** Multivariable Calculus, Linear Algebra, Probability Theory, Mathematical Statistics, Algorithms, Computational Complexity, Computer Vision

Experience

Computer Vision Student Researcher (Near Earth Autonomy)

January 2026 – present

- Investigated and implemented improvement modules on YOLOv8-seg, leading to a 4% improvement in mAP for monocular runway segmentation from simulated fixed-wing aircraft over baseline YOLO
- Developed a joint estimator for aircraft yaw and cross-track distance by optimizing segmentation masks and fusing predictions using extended Kalman filtering, achieving a maximum of 0.4 degrees of yaw error across test suite of 10+ flights
- Built a per-frame geometry-based estimator of cross-track distance using bird's-eye view transforms and Kalman filtering, achieving an average of 1.22 meters of error across test suite of 120+ landing sequences

Machine Learning Validation Intern (General Motors / Cruise)

May 2026 – August 2026

- Developed AI agent skills to search national-scale road map with natural language descriptions using Model Workspace Protocol (MWP) design and Google BigQuery. Achieved over 70% precision during testing on a suite of over twenty varied descriptions and reduced development time of specialized geometry-based map queries from over one week to approximately one hour
- Prototyped AI-based pipeline for categorization of type and severity of inclement weather using a lightweight CNN, achieving up to 90% precision and recall on a curated testing set of over 100 drives
- Led cross-team modernization of perception stack KPIs and developed confusion matrices, precision-recall curves, and top mover analyses for over 1000 drive simulations

Undergraduate Research Fellow (Boston College)

January 2022 – August 2025

- Developed UNet-based machine learning pipelines for cell segmentation in electron microscopy imagery over multiple projects. Fine-tuned and scaled models during deployment to generate segmentation maps on terabytes of data. Advised by Professor Donglai Wei and collaborated with labs at USC, Columbia University, and UMichigan to publish multiple papers in high-profile biology journals (Nature, Cell)
- Organized and facilitated lab's onboarding program and instructed new team members on MLOps tools (Numpy, PyTorch, PyTorch Connectomics, Slurm) through weekly in-person meetings and coding exercises
- Prepared weekly presentations for multiple interdisciplinary groups of researchers. Presented on machine learning pipeline development to undergraduate, graduate, and post-graduate collaborators

Undergraduate Teaching Assistant (Boston College)

August 2022 – May 2023

- Instructed over 40 students on computer hardware (Verilog) and foundational machine learning packages (Numpy, OpenCV, Pytorch) across two semesters. Held weekly office hours for two courses (Computer Organization and Lab, Biomedical Image Analysis)
- Graded problem sets and labs, delivering reports on student performance to supervising professor

General Service Server (Boston College Dining Services)

May 2022 – August 2022

- Coordinated with group of ten for setup and breakdown of venue for school orientation, serving over 8000 guests over ten weeks
- Delivered excellent customer service with an amenable and welcoming attitude, promptly clearing tables for expedited table turnaround time and improving customer experience

Projects

Visual Odometry and SLAM in Game Environments

January 2026 – May 2026

- Benchmarked classical and learning-based visual odometry and SLAM methods (ORB-SLAM3, MAST3R-SLAM, DPVO, TartanVO) on 572 trajectories across 31 photorealistic video game scenes from the OmniWorld dataset. Built a unified pipeline for data conversion and trajectory export, and used Sim(3)-aligned ATE and RPE to evaluate global consistency and local drift
- Investigated DPVO's patch selection under fast camera motion and applied region masking to remove the sky and player model from tracking, which reduced translational RPE by up to 19% on a challenging scene

Open-Vocabulary Scene Understanding for 3D Gaussian Splatting

September 2025 – December 2025

- Developed a 3D Gaussian splatting tool based on OpenGaussian and StreetGaussian for zero-shot semantic understanding. Used 2D segmentation masks from Segment Anything to facilitate learning of a discrete codebook of open vocabulary language embeddings and utilized cosine distance as a guiding metric for answering natural language queries

3D Cell Segmentation and Center-line Extraction

May 2025 – August 2025

- Developed cell segmentation, center-line fitting, and signal extraction pipelines using PyTorch Connectomics and Segment Anything to study gene regulation dynamics. Completed in collaboration with the Spatial Biodynamics Lab at the University of Michigan. Created over 5000 cell instance segmentation and center-line prediction pairs with precision/recall scores of 0.6/0.7 during deployment
- Optimized model hyperparameters using algorithms built on Scipy and the Optuna sampling framework, increasing model recall by 10%

Large-scale 3D Segmentation in Electron Microscopy Volumes

May 2024 – March 2025

- Created a machine learning pipeline based on the 3D-UNet architecture to study cell distribution in nervous systems of model organisms. Created instance segmentation maps of vesicles on large-scale electron microscopy volumes of *Hydra vulgaris* using tools from the PyTorch Connectomics library. Completed in collaboration with Prof. Rafael Yuste's NeuroTechnology Lab at Columbia University
- Implemented human-in-the-loop design principles during model training to expedite data annotation over three rounds of inference and annotation, increasing labeling throughput by over 10 times
- Automated model deployment (Slurm, sequential job management, bash scripting) on over 1TB of volumetric data on a HPC, producing 40000 instance segmentations with 98% precision

Real-time Microscopy Object Detection

January 2023 – September 2023

- Developed a UNet model for object detection in STEM images using Pytorch to study electromagnetic spin in ferroelectric materials. Generated over 100 heatmaps of merons detections from greyscale EM imagery, achieving over 70% precision and recall during testing at a 1Hz rate. Completed in collaboration with the Shao Electron Microscopy Group at USC
- Published and presented a poster at the Microscopy and Microanalysis conference in 2024. Fielded questions over two hours during conference and made over 400 direct impressions by January 2026

Publications (* indicates shared first author)

Lirong Zheng, ..., Michael Lin, et al (2025). *Scalable and multiplexed recorders of gene regulation dynamics across weeks*. Nature. doi.org/10.1038/s41586-026-10156-9.

Shulin Zhang, ..., Michael Lin, et al (2025). *Ultrastructural reconstruction of the endodermal nerve net of Hydra vulgaris*. Current Biology. doi.org/10.1016/j.cub.2025.10.001.

Michael Lin*, et al. *VesicleEM: A Comprehensive Vesicle Analysis Toolbox for Volumetric Electron Microscopy*. doi.org/10.1371/journal.pcbi.1013499

Michael Lin, et al (2024). *Real-Time Point of Interest Segmentation for Electron Microscopy Images via Machine Learning*. Microscopy and Microanalysis. doi.org/10.1093/mam/ozae044.215.

Technical Skills

Languages: English (native proficiency), Mandarin Chinese (conversational proficiency)

Programming Languages: Python, C, C++, Java, R, SQL

Tools: Pytorch, OpenCV, Numpy, PIL, Scikit, Matplotlib, Scipy, Git, Slurm, Github, Linux, Jupyter, Optuna